Design and Simulation of a Parameterized Traffic Light Controller Using Verilog HDL

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Abstract

This project presents the design and behavioral simulation of a parameterized Traffic Light Controller using Verilog Hardware Description Language (HDL). The controller is based on a Moore Finite State Machine (FSM) that efficiently manages traffic flow at a four-way road intersection by controlling the sequence and timing of traffic signals. The design is fully synchronous and utilizes a single system clock with a clock-enable mechanism to ensure reliable and predictable operation.

The proposed controller incorporates several practical traffic management features, including pedestrian crossing requests, emergency vehicle override, vehicle detection through traffic sensors, and a night mode with blinking yellow signals for low-traffic conditions. To improve the reliability of external inputs, dual flip-flop synchronization and counter-based debounce logic are employed for all asynchronous signals.

The design is implemented in Verilog HDL and functionally verified through behavioral simulation using EDA Playground with the Icarus Verilog 12.0 simulator. Simulation waveforms are analyzed using the EPWave waveform viewer to validate the correct operation of the traffic light sequences and special operating modes. The modular and parameterized architecture makes the controller easy to understand, verify, and extend for future traffic management applications.

Keywords: Traffic Light Controller, Verilog HDL, Finite State Machine (FSM), Behavioral Simulation, Traffic Signal Control, Icarus Verilog, EDA Playground.

Table of Contents

List of Tables

List of Figures

Abbreviations

Abbreviation	Description
CORDIC	Coordinate Rotation Digital Computer
HGH-CORDIC	High-Radix Generalized Hyperbolic CORDIC
DSP	Digital Signal Processing
FPGA	Field Programmable Gate Array
HDL	Hardware Description Language
VHDL	Very High Speed Integrated Circuit Hardware Description Language
RTL	Register Transfer Level
LUT	Look-Up Table
WNS	Worst Negative Slack
TNS	Total Negative Slack
AGC	Automatic Gain Control
ASIC	Application Specific Integrated Circuit
ALU	Arithmetic Logic Unit
ITER	Iterations

CHAPTER 1

Introduction

1.1 Introduction

Rapid urbanization and the continuous increase in the number of vehicles on roads have made efficient traffic management an essential requirement for modern transportation systems. Conventional traffic signal systems often operate using fixed timing intervals, which may not efficiently handle varying traffic conditions, emergency situations, or pedestrian crossing requests. Inefficient traffic signal control can lead to increased traffic congestion, longer waiting times, higher fuel consumption, and unnecessary environmental pollution. Therefore, the development of intelligent and reliable traffic control systems has become an important area of research in digital system design and embedded applications.

Traffic Light Controllers are widely used at road intersections to regulate vehicle movement and ensure the safe passage of both vehicles and pedestrians. A well-designed traffic controller coordinates traffic signals by providing appropriate green, yellow, and red light durations while preventing conflicting traffic movements. Modern traffic management systems also incorporate additional features such as pedestrian crossing requests, emergency vehicle priority, vehicle detection sensors, and night mode operation to improve traffic efficiency and road safety.

Digital hardware provides an effective platform for implementing traffic control systems due to its speed, reliability, and deterministic operation. Hardware Description Languages (HDLs) such as Verilog HDL enable designers to model, simulate, and verify complex digital circuits before hardware implementation. Compared to software-based controllers, hardware-based controllers offer predictable timing, faster response, and greater reliability, making them suitable for real-time traffic management applications.

One of the most widely used techniques for designing sequential digital controllers is the Finite State Machine (FSM). An FSM represents system behavior as a collection of states and transitions, allowing complex control logic to be implemented in a structured and efficient manner. Among the different FSM models, the Moore Finite State Machine is particularly suitable for traffic light control because its outputs depend only on the current state, resulting in stable and predictable traffic signal operation. This approach simplifies controller design while ensuring safe signal transitions under different operating conditions.

Modern traffic controllers must also address practical challenges associated with external input signals. Inputs such as pedestrian push buttons, emergency switches, and vehicle sensors are asynchronous with respect to the system clock and may introduce metastability or false triggering due to switch bouncing. To improve system reliability, synchronization circuits and debounce logic are commonly employed to ensure that only valid input signals are processed by the controller.

This project presents the design and behavioral simulation of a parameterized Traffic Light Controller using Verilog Hardware Description Language (HDL). The controller is based on a Moore Finite State Machine that manages traffic flow at a four-way road intersection while supporting practical operating modes including pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation. The design employs a single system clock with a clock-enable mechanism to maintain fully synchronous operation and incorporates synchronization and debounce circuits for reliable handling of external inputs.

The proposed design is developed using a modular architecture in Verilog HDL, where individual modules perform specific functions such as clock enable generation, timer control, input synchronization, debounce logic, and finite state machine control. This modular approach improves code readability, simplifies verification, and allows future enhancements without affecting the overall system architecture.

The functionality of the proposed controller is verified through behavioral simulation using EDA Playground with the Icarus Verilog 12.0 simulator. Simulation waveforms generated using the EPWave waveform viewer confirm the correct sequence of traffic light transitions under normal operation as well as during pedestrian requests, emergency conditions, and night mode operation. The parameterized architecture provides flexibility for modifying signal timing and adapting the controller to different traffic management requirements while maintaining a simple and reliable hardware design.

1.2 Background and Motivation

Efficient traffic management is one of the major challenges faced by modern cities due to the continuous growth in the number of vehicles on the road. Traffic congestion not only increases travel time but also leads to higher fuel consumption, environmental pollution, and an increased risk of road accidents. Traffic signal control systems play a vital role in regulating vehicle movement at road intersections and ensuring the safe and orderly flow of traffic. Therefore, the development of reliable and efficient traffic light controllers has become an important area of research in digital system design.

Traditional traffic signal systems generally operate using fixed timing schedules regardless of the actual traffic conditions. Although these systems are simple to implement, they often result in inefficient utilization of road infrastructure during varying traffic conditions. They usually do not provide support for special situations such as emergency vehicle priority, pedestrian crossing requests, or reduced traffic flow during nighttime. As a result, modern traffic management systems require more flexible and intelligent control mechanisms that can adapt to different operating scenarios while maintaining safe traffic operation.

Digital hardware offers an effective solution for implementing traffic controllers due to its deterministic behavior, high processing speed, and reliability. Hardware Description Languages (HDLs) such as Verilog HDL enable designers to model, simulate, and verify digital systems before hardware implementation. Compared to software-based approaches, hardware-based controllers provide predictable timing characteristics and are well suited for real-time control applications where accurate signal sequencing is essential.

Finite State Machines (FSMs) are widely used in the design of digital controllers because they provide a structured method for implementing sequential logic. In a traffic light controller, the FSM defines the sequence of traffic signal states and determines the transitions between them based on timing intervals and external inputs. The Moore FSM model is particularly suitable because the output signals depend only on the current state, ensuring stable and glitch-free operation of the traffic lights throughout each signal cycle.

In practical traffic control systems, external inputs such as pedestrian push buttons, vehicle sensors, and emergency signals are asynchronous with respect to the system clock. These signals may introduce metastability or generate multiple unwanted transitions due to switch bouncing. To improve system reliability, synchronization circuits and debounce logic are incorporated into the controller to ensure that only valid input signals are processed. This enhances the robustness of the system and prevents incorrect state transitions.

The motivation for this project is to develop a flexible and modular traffic light controller that addresses the limitations of conventional fixed-time systems while remaining simple to implement and verify. The proposed controller incorporates several practical features, including pedestrian crossing requests, emergency vehicle override, vehicle detection support, and night mode operation with blinking signals. The design follows a modular architecture consisting of separate modules for clock enable generation, timer control, debounce and synchronization logic, and finite state machine control.

The controller is implemented using Verilog Hardware Description Language (HDL) and verified through behavioral simulation using EDA Playground with the Icarus Verilog 12.0 simulator. The modular and parameterized design improves scalability, simplifies future enhancements, and provides an effective learning platform for digital system design, finite state machine implementation, and traffic management applications.

1.2.1 Objectives of the Project

Efficient traffic management is essential for ensuring smooth vehicle movement and improving road safety at busy intersections. Conventional traffic signal systems with fixed timing are often unable to respond effectively to varying traffic conditions, pedestrian requests, or emergency situations. Digital hardware-based controllers provide a reliable and efficient solution by offering deterministic operation, high processing speed, and flexible control logic. In this context, the main objective of this project is to design and simulate a parameterized Traffic Light Controller using Verilog Hardware Description Language (HDL) based on a Moore Finite State Machine (FSM).

The specific objectives of this project are as follows:

- To study the working principles of traffic signal control systems and Finite State Machine (FSM) based digital controllers.
- To design a parameterized traffic light controller capable of managing traffic flow at a four-way road intersection.
- To implement pedestrian crossing requests, emergency vehicle override, vehicle detection support, and night mode operation within the controller.
- To develop reliable synchronization and debounce circuits for handling asynchronous external inputs.
- To implement the complete controller using Verilog Hardware Description Language with a modular design approach.
- To verify the functionality of the proposed design through behavioral simulation using EDA Playground with the Icarus Verilog simulator.
- To validate the correct sequence of traffic signal transitions under normal and special operating conditions using simulation waveforms.

By achieving these objectives, the project demonstrates a reliable and modular hardware-based traffic light control system suitable for digital design applications. The parameterized architecture improves flexibility, simplifies future enhancements, and provides an effective platform for understanding finite state machine based control systems.

1.2.2 Organization of the Project

This report is organized into several sections to systematically present the design, implementation, and verification of the proposed Traffic Light Controller. Each section describes a specific stage of the project, beginning with the theoretical background and progressing through system design, implementation, simulation, and final conclusions.

- **Literature Review**

  This section presents an overview of existing traffic light control systems and digital controller designs. Previous research on FSM-based traffic controllers, intelligent traffic management techniques, and hardware-based implementations is discussed to understand their advantages and limitations.

- **Background**

  This section explains the theoretical concepts required for the project, including digital logic design, Finite State Machines (FSMs), Moore machine architecture, traffic signal control principles, and synchronization techniques used for reliable digital systems.

- **Methodology**

  This section describes the overall design methodology followed in the project. It explains the development of the modular architecture, FSM design, timer logic, synchronization and debounce circuits, Verilog HDL implementation, and behavioral simulation process.

- **Proposed Work**

  This section presents the proposed parameterized Traffic Light Controller architecture. It explains the functional modules, state transitions, traffic signal sequencing, pedestrian request handling, emergency vehicle override, vehicle detection logic, and night mode operation.

- **Implementation**

  This section discusses the implementation of the controller using Verilog Hardware Description Language. It describes the individual RTL modules, top-level integration, parameter configuration, and overall design structure.

- **Results and Discussions**

  This section presents the behavioral simulation results obtained using EDA Playground with the Icarus Verilog simulator. Simulation waveforms are analyzed to verify the correct operation of normal traffic sequencing, pedestrian mode, emergency override, vehicle detection, and night mode functionality.

- **Conclusions and Recommendations**

  This section summarizes the overall achievements of the project and discusses the effectiveness of the proposed traffic light controller. It also presents possible future enhancements, such as adaptive traffic control, FPGA implementation, IoT integration, and AI-based traffic optimization.

CHAPTER 2

Literature Survey

2.1 Introduction

Efficient traffic management has become increasingly important due to rapid urbanization and the growing number of vehicles on road networks. Conventional traffic signal systems with fixed timing schedules often fail to adapt to changing traffic conditions, resulting in unnecessary delays, increased fuel consumption, and traffic congestion. To address these challenges, researchers have proposed various digital traffic control systems that improve traffic flow, enhance road safety, and support intelligent transportation infrastructure.

Finite State Machines (FSMs) are widely used in the design of digital traffic light controllers because they provide a systematic approach for implementing sequential control logic. FSM-based controllers ensure predictable state transitions, reliable signal sequencing, and simplified hardware implementation. When combined with Hardware Description Languages (HDLs) such as Verilog HDL, FSM-based designs can be efficiently modeled, simulated, verified, and later implemented on programmable logic devices.

Modern traffic light controllers extend beyond simple fixed-time operation by incorporating features such as pedestrian crossing requests, emergency vehicle priority, vehicle detection sensors, and night mode operation. These intelligent features improve the flexibility of traffic management systems while maintaining safe operation under different traffic conditions. Additionally, synchronization and debounce techniques are commonly employed to ensure reliable processing of asynchronous external inputs such as push buttons and sensor signals.

Recent developments in digital system design have encouraged the use of modular and parameterized hardware architectures. Modular designs simplify verification, improve code reusability, and allow future enhancements without modifying the complete system. Parameterized controllers also provide flexibility in adjusting traffic signal timings according to different intersection requirements.

This project follows a modular FSM-based approach for designing a parameterized Traffic Light Controller using Verilog Hardware Description Language. The controller supports normal traffic operation, pedestrian requests, emergency vehicle override, vehicle detection, and night mode functionality. Behavioral simulation is performed using EDA Playground with the Icarus Verilog simulator to verify the correctness of the controller under various operating conditions.

2.2 Review of Prior Research

Early traffic light control systems primarily relied on fixed-time signal scheduling, where each traffic direction received predetermined green, yellow, and red signal durations. Although these systems were simple to implement, they were unable to respond efficiently to varying traffic densities, resulting in increased waiting times and reduced traffic throughput.

Researchers later introduced Finite State Machine (FSM) based traffic controllers to improve system reliability and simplify digital hardware implementation. Moore and Mealy FSM models have been extensively used to implement traffic signal sequencing because they provide deterministic state transitions and predictable control behavior. Among these, Moore FSMs are commonly preferred since the output signals depend only on the current state, reducing the possibility of output glitches during state transitions.

Several studies have proposed intelligent traffic controllers that integrate pedestrian crossing support, emergency vehicle priority, and vehicle detection mechanisms. These additional features improve road safety while enabling more efficient utilization of road infrastructure. Emergency override mechanisms allow priority access for emergency vehicles, whereas pedestrian request handling provides safer road crossing without disrupting the overall traffic sequence.

Modern hardware implementations increasingly utilize Verilog HDL due to its ability to model complex sequential systems with high reliability. Modular Verilog designs improve maintainability by separating the controller into functional units such as finite state machines, timer modules, synchronization circuits, debounce logic, and top-level integration. This modular architecture simplifies debugging, testing, and future expansion of the system.

Recent research has also emphasized the importance of reliable input processing in digital controllers. Since pedestrian buttons and external sensors generate asynchronous signals, synchronization and debounce circuits are incorporated to eliminate metastability and switch bouncing effects. These techniques improve system robustness and ensure that only valid input events influence the controller operation.

The proposed Traffic Light Controller builds upon these concepts by implementing a parameterized Moore FSM architecture together with dedicated timer, clock-enable, synchronization, and debounce modules. The design combines modular hardware architecture with practical traffic management features, resulting in a reliable and easily extensible digital traffic control system suitable for behavioral simulation and future hardware implementation.

2.3 Identified Research Gaps

From the literature survey, the following research gaps have been identified:

- Many conventional traffic light controllers operate using fixed timing schedules and are unable to adapt to changing traffic conditions or special operational requirements.
- Several existing traffic control systems focus only on basic traffic signal sequencing and do not incorporate features such as pedestrian crossing requests, emergency vehicle priority, vehicle detection, and night mode operation.
- In many digital controller designs, asynchronous inputs from push buttons and sensors are not properly synchronized or debounced, which can result in unreliable operation and incorrect state transitions.
- Existing implementations often use monolithic designs, making the controller difficult to understand, verify, modify, and extend for future

applications.

- There is a need for a modular and parameterized Verilog HDL implementation that supports multiple traffic management features while remaining easy to simulate, verify, and adapt to different traffic intersection requirements.

2.4 Summary

This chapter reviewed the existing approaches for designing digital traffic light controllers and highlighted the importance of Finite State Machine (FSM) based architectures for reliable traffic signal control. Previous studies have demonstrated the effectiveness of FSM-based controllers in providing deterministic operation and simplified hardware implementation. However, many existing designs either lack practical traffic management features or do not adequately address the reliable handling of asynchronous external inputs.

Based on these observations, this project presents the design and behavioral simulation of a parameterized Traffic Light Controller using Verilog Hardware Description Language. The proposed controller integrates pedestrian request handling, emergency vehicle override, vehicle detection support, night mode operation, and reliable synchronization with debounce logic within a modular architecture. Behavioral simulation is performed using EDA Playground with the Icarus Verilog simulator to verify the correct operation of the controller under various traffic conditions. The modular design also provides flexibility for future enhancements and potential hardware implementation.

CHAPTER 3

Review of Prior Research

3.1 Introduction

Efficient traffic management has become an important research area due to the rapid increase in urban population and vehicle density. Conventional traffic signal systems based on fixed timing schedules are often unable to efficiently manage varying traffic conditions, resulting in increased congestion, longer waiting times, higher fuel consumption, and reduced road safety. To overcome these limitations, researchers have proposed several intelligent traffic control techniques that improve the efficiency and reliability of traffic signal operation.

Digital traffic light controllers implemented using Hardware Description Languages (HDLs) have gained significant attention because of their high speed, reliability, and deterministic operation. Verilog HDL enables designers to model, simulate, and verify digital systems before hardware implementation, making it an effective choice for developing real-time traffic control systems. Compared to software-based controllers, hardware-based designs provide predictable timing characteristics and are better suited for safety-critical applications such as traffic signal control.

Finite State Machines (FSMs) are widely used in digital traffic controller design because they provide a systematic approach for implementing sequential control logic. Moore FSMs are particularly suitable for traffic light control applications since the output signals depend only on the current state, ensuring stable and glitch-free traffic signal transitions. Researchers have also incorporated additional features such as pedestrian crossing requests, emergency vehicle priority, vehicle detection sensors, and night mode operation to improve traffic efficiency and road safety.

Recent developments in digital design have emphasized modular and parameterized hardware architectures that simplify verification, improve maintainability, and allow future system expansion. Synchronization circuits and debounce logic are also widely adopted to ensure reliable processing of asynchronous inputs generated by pedestrian push buttons and vehicle sensors.

This chapter reviews previous research related to traffic light control systems, FSM-based digital controller design, Verilog HDL implementation techniques, and intelligent traffic management features. The review provides an understanding of existing approaches, their advantages, limitations, and the motivation behind the proposed parameterized Traffic Light Controller developed in this project.

3.2 Finite State Machine (FSM) Based Traffic Light Controller

A Finite State Machine (FSM) is one of the most widely used models for designing sequential digital systems. It represents the behavior of a system as a finite number of states, where transitions between states occur based on input conditions and timing requirements. FSMs provide a structured and reliable approach for implementing control logic, making them particularly suitable for applications such as traffic light controllers.

In a traffic light controller, each state corresponds to a specific combination of traffic signals for different directions of the intersection. The controller continuously transitions between these states according to predefined timing intervals while responding to external events such as pedestrian requests, emergency vehicle signals, vehicle detection sensors, and night mode operation. This state-based approach ensures safe and orderly traffic movement by preventing conflicting traffic signals.

The proposed controller employs a Moore Finite State Machine, in which the output signals depend only on the current state of the system. Since the outputs are independent of instantaneous input changes, Moore FSMs provide stable and glitch-free signal transitions, making them highly suitable for safety-critical applications such as traffic signal control.

The FSM implemented in this project consists of multiple operating states that regulate the sequence of green, yellow, and red signals for each traffic direction. State transitions are controlled using a timer module together with synchronized external inputs, ensuring predictable and reliable operation throughout the traffic cycle. The modular FSM architecture also simplifies verification and allows future modifications with minimal changes to the overall design.

3.3 Moore FSM Architecture for Traffic Signal Control

The Moore Finite State Machine forms the core of the proposed Traffic Light Controller. In a Moore FSM, the output signals are determined solely by the current state, while state transitions depend on the system clock, timer values, and validated external inputs. This architecture provides deterministic behavior and eliminates unwanted output glitches that may occur due to asynchronous input changes.

The proposed controller manages traffic flow through a sequence of predefined states representing different traffic signal combinations. During normal operation, the controller cycles through green, yellow, and red signal phases for each direction according to configurable timing parameters. Additional operating modes are incorporated to improve the functionality of the controller.

Pedestrian requests are processed by temporarily modifying the normal traffic sequence to provide safe crossing opportunities. Emergency vehicle override gives immediate priority by forcing the controller into a predefined safe traffic state. Vehicle detection inputs can be used to improve traffic flow by responding to traffic demand, while the night mode enables blinking signal operation during periods of low traffic.

To ensure reliable operation, all asynchronous external inputs pass through synchronization and debounce circuits before being processed by the finite state machine. A clock-enable module and timer module generate the required timing intervals while maintaining a fully synchronous design using a single system clock.

The modular architecture adopted in this project separates the controller into functional blocks including the clock-enable generator, timer module, synchronization and debounce logic, finite state machine controller, and top-level integration module. This design approach improves readability, simplifies debugging, and makes the controller easily scalable for future traffic management applications.

3.4 FSM-Based Traffic Controller Architectures

Finite State Machine (FSM) based architectures are widely adopted for designing digital traffic light controllers because they provide a structured approach for implementing sequential control systems. Each state represents a specific traffic signal configuration, while transitions between states are determined by timing intervals and external input conditions. This approach simplifies controller design and ensures reliable traffic signal operation.

Modern FSM-based traffic controllers extend conventional fixed-time signal systems by incorporating intelligent features that improve traffic management and road safety. These features include pedestrian crossing requests, emergency vehicle priority, vehicle detection, and night mode operation. By integrating these capabilities into the finite state machine, the controller can respond effectively to different traffic situations while maintaining safe signal transitions.

A modular FSM architecture further improves the design by separating the controller into independent functional modules such as the finite state machine, timer, clock-enable generator, synchronization and debounce circuits, and top-level integration. This modular approach improves readability, simplifies debugging, and allows future enhancements without significantly modifying the overall system architecture.

The proposed Traffic Light Controller follows a parameterized Moore FSM architecture that provides configurable timing intervals and supports multiple operating modes. The modular implementation makes the controller scalable and suitable for educational, research, and future hardware implementation purposes.

3.5 Hardware Implementation Considerations

Designing a reliable digital traffic light controller requires careful consideration of timing, synchronization, modularity, and input reliability. Since traffic signal control is a real-time application, all signal transitions must occur in a predictable and deterministic manner. Therefore, the proposed controller follows a fully synchronous design using a single system clock together with a clock-enable mechanism to control timing operations.

External inputs such as pedestrian push buttons, emergency signals, and vehicle detection sensors are asynchronous with respect to the system clock. If these inputs are directly connected to the finite state machine, they may cause metastability or multiple unwanted transitions due to switch bouncing. To improve reliability, synchronization and debounce circuits are incorporated before processing these signals within the controller.

The controller is implemented using Verilog Hardware Description Language (HDL) with a modular design approach. Individual modules perform specific functions, including clock-enable generation, timer control, input synchronization, debounce logic, finite state machine control, and top-level integration. This modular organization improves code maintainability, simplifies verification, and allows individual modules to be tested independently.

Behavioral simulation is carried out using EDA Playground with the Icarus Verilog 12.0 simulator, while the generated waveforms are analyzed using EPWave. Simulation verifies the correct operation of the controller under normal traffic sequencing, pedestrian request handling, emergency vehicle override, vehicle detection, and night mode operation. The modular and parameterized architecture also provides flexibility for future hardware implementation and additional traffic management features.

3.5.1 Hardware Architecture

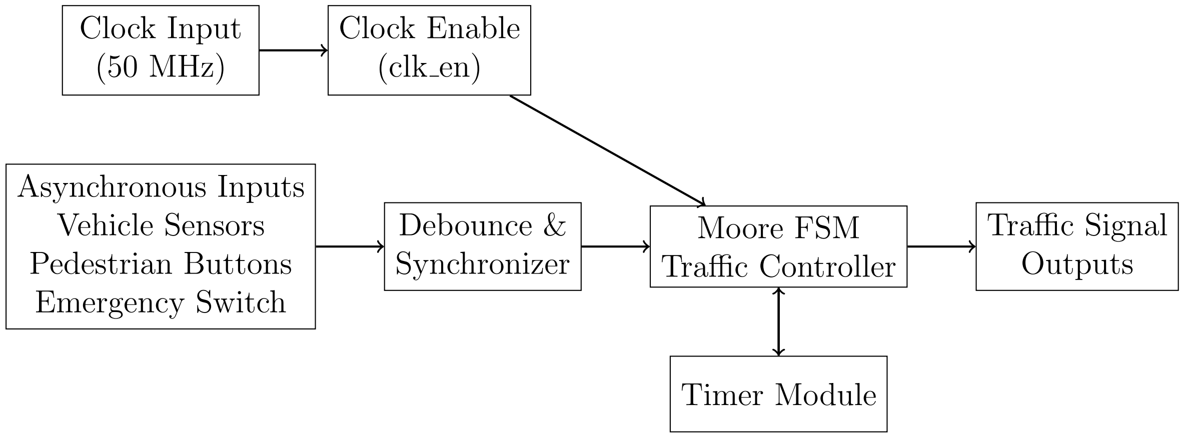

Figure 3.1: Hardware Architecture of the Proposed Traffic Light Controller

The proposed Traffic Light Controller follows a modular hardware architecture implemented using Verilog Hardware Description Language (HDL). A single 50 MHz system clock serves as the primary timing source for the entire design. The `clk_en` module generates a periodic 10 Hz clock-enable pulse that controls the timing of traffic signal transitions while maintaining a fully synchronous design.

External inputs, including vehicle detection sensors, pedestrian push buttons, and emergency override switches, are asynchronous with respect to the system clock. These inputs are first processed by the debounce and synchronization module to eliminate switch bouncing and reduce the possibility of metastability. The validated input signals are then provided to the Moore Finite State Machine (FSM), which acts as the central controller of the system.

The timer module works together with the FSM by monitoring the duration of each traffic state and generating timeout events required for state transitions. Based on the timer status and external input conditions, the FSM determines the next operating state and generates the appropriate traffic signal outputs.

The controller supports multiple operating modes including normal traffic sequencing, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation. The output signals generated by the FSM control the traffic light clusters and pedestrian indicators while ensuring safe and conflict-free signal transitions. The modular organization of the design simplifies verification, improves maintainability, and allows future enhancements without modifying the overall system architecture.

3.6 Summary

This chapter reviewed existing research related to digital traffic light control systems and Finite State Machine (FSM) based controller architectures. Conventional traffic controllers, FSM-based implementations, modular digital design techniques, and intelligent traffic management features were examined to understand their advantages and limitations.

The review highlighted the importance of reliable state transition mechanisms, synchronization of asynchronous inputs, modular hardware architecture, and configurable timing for developing efficient traffic management systems. It also emphasized the growing need for practical traffic control features such as pedestrian request handling, emergency vehicle priority, vehicle detection, and night mode operation.

The concepts discussed in this chapter provide the foundation for the proposed parameterized Traffic Light Controller presented in this work. The following chapter describes the overall system methodology, design approach, module organization, and implementation of the controller using Verilog HDL, followed by behavioral simulation and functional verification.

CHAPTER 4

Proposed Architecture for Traffic Light Controller

4.1 Introduction

Efficient traffic management is essential for ensuring smooth vehicle movement, minimizing congestion, and improving road safety at road intersections. Conventional traffic light systems generally operate using fixed timing schedules and are unable to respond effectively to varying traffic conditions, pedestrian requests, or emergency situations. As a result, there is a need for intelligent digital controllers that provide reliable operation while supporting multiple traffic management features.

Finite State Machine (FSM) based controllers provide an efficient solution for implementing sequential traffic control logic. By representing the operation of the traffic controller as a sequence of predefined states, FSMs ensure deterministic behavior, simplified hardware implementation, and reliable signal transitions. A Moore Finite State Machine is particularly suitable because its outputs depend only on the current state, resulting in stable traffic signal operation throughout each traffic cycle.

The proposed Traffic Light Controller is implemented using Verilog Hardware Description Language (HDL) with a modular architecture consisting of separate modules for clock-enable generation, timer control, debounce and synchronization logic, finite state machine control, and top-level integration. The controller supports normal traffic sequencing together with pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation. Behavioral simulation is performed using EDA Playground with the Icarus Verilog simulator to verify the functionality of the proposed design.

4.2 Proposed Traffic Light Control Algorithm

The proposed controller operates as a Moore Finite State Machine that continuously cycles through a predefined sequence of traffic signal states. State transitions are controlled using configurable timing intervals together with validated external input signals obtained from pedestrian push buttons, vehicle sensors, emergency switches, and night mode selection.

During normal operation, the controller alternates traffic flow between the North–South and East–West directions while maintaining appropriate yellow transition intervals and all-red safety intervals. When a pedestrian request is detected, the controller temporarily modifies the normal sequence to provide a dedicated pedestrian crossing period. If an emergency signal is received, the controller immediately enters a safe emergency state by forcing all traffic directions to red. During low-traffic conditions, the controller can enter night mode, where blinking signals are generated according to the selected operating mode.

The controller utilizes synchronized and debounced external inputs to ensure reliable operation while preventing false triggering caused by switch bouncing or asynchronous signal transitions. A timer module determines the duration of each traffic state and generates timeout signals that control state transitions within the finite state machine.

4.2.1 Pseudocode

```
Input:
    Clock
    Vehicle Sensors
    Pedestrian Button
    Emergency Switch
    Night Mode Switch

Output:
    North-South Signals
```

```
    East-West Signals
    Pedestrian Indicators

1. Initialize system and reset FSM.
2. Generate periodic clock-enable pulse.
3. Synchronize and debounce all external inputs.
4. Enter normal traffic operation.
5. Repeat continuously:
     • If emergency signal is active:
           Enter Emergency State
     • Else if night mode is enabled:
           Enter Night Mode
     • Else
           Execute normal traffic sequence:
               North-South Green
               North-South Yellow
               All Red
               Pedestrian Walk (if requested)
               East-West Green
               East-West Yellow
               All Red
     • Update timer for current state.
     • Change state after timer expires.
6. Repeat until system reset.
```

4.3 Proposed Traffic Light Controller Architecture

The proposed Traffic Light Controller is designed using a modular architecture in Verilog Hardware Description Language (HDL). Each module performs a specific function, making the overall design easier to understand, verify, and maintain. The major components of the proposed architecture are as follows:

- **Clock Enable Module** (`clk_en`)
  Generates a periodic clock-enable pulse from the system clock to control the timing of traffic signal transitions while maintaining a fully synchronous design.

- **Debounce and Synchronization Module** (`debounce_sync`)
  Processes asynchronous external inputs such as pedestrian push buttons, vehicle sensors, emergency switches, and night mode switches. The module removes switch bouncing effects and synchronizes the inputs with the system clock.

- **Moore Finite State Machine Module** (`traffic_fsm`)
  Acts as the central controller of the system by managing the sequence of traffic light states. It controls normal traffic operation, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation.

- **Timer Module** (`timer`)
  Measures the duration of each traffic state and generates timeout signals required for state transitions within the finite state machine.

- **Top-Level Integration Module** (`top`)
  Integrates all functional modules into a complete Traffic Light Controller and generates the final traffic signal outputs for the North–South direction, East–West direction, and pedestrian indicators.

4.4 Algorithm Flow

The overall development flow of the proposed Traffic Light Controller is summarized below.

The development process begins by defining the functional requirements of the Traffic Light Controller, including normal traffic operation, pedestrian crossing, emergency vehicle override, vehicle detection, and night mode functionality. A Moore Finite State Machine (FSM) is then designed to manage the

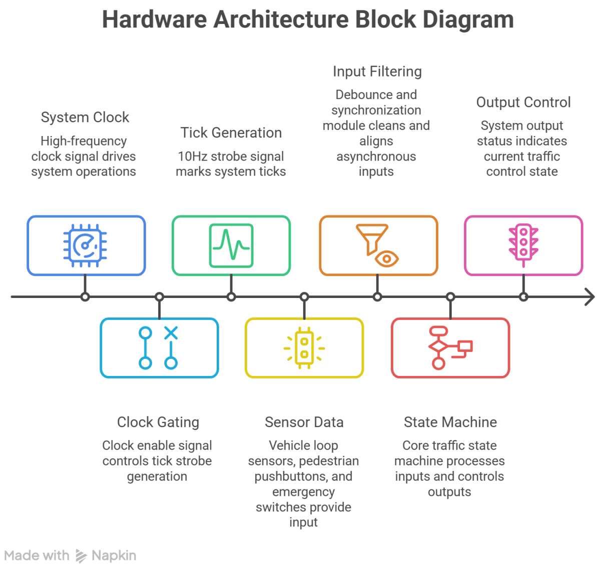

Figure 4.1: Hardware Architecture Block Diagram

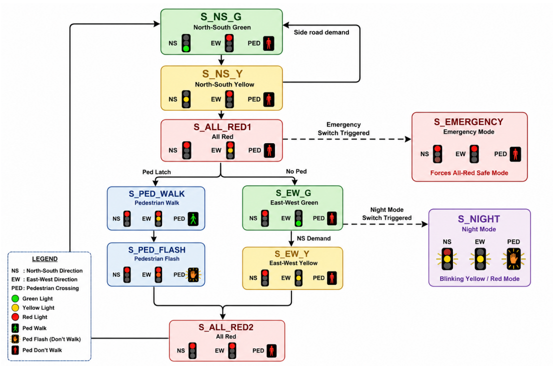

Figure 4.2: Moore Finite State Machine Diagram

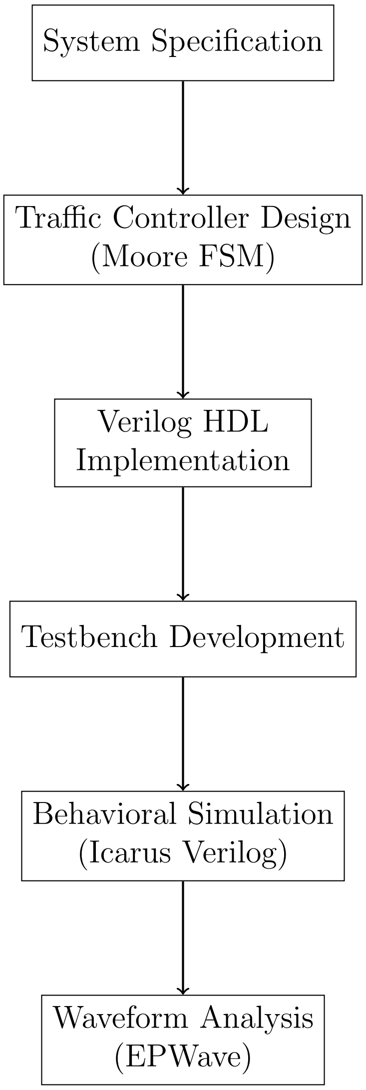

Figure 4.3: Development Flow of the Proposed Traffic Light Controller

sequence of traffic signal states. The controller is implemented using Verilog Hardware Description Language (HDL) with a modular architecture consisting of clock-enable generation, timer control, synchronization and debounce logic, finite state machine control, and top-level integration. A dedicated testbench is developed to verify the controller under different operating conditions. Behavioral simulation is performed using the Icarus Verilog simulator in EDA Playground, and the generated waveforms are analyzed using EPWave to validate the correct operation of the traffic signal sequences and special operating modes.

4.5 Summary

This chapter presented the proposed architecture of the Traffic Light Controller based on a Moore Finite State Machine implemented using Verilog HDL. The modular design incorporates clock-enable generation, timer control, synchronization and debounce logic, pedestrian request handling, emergency vehicle override, vehicle detection, and night mode operation. The controller was functionally verified through behavioral simulation using Icarus Verilog, and the generated waveforms confirmed the correct operation of all traffic control modes. The next chapter presents the implementation details and discusses the simulation results obtained from the proposed design.

CHAPTER 5

Results and Discussions

5.1 Introduction

This chapter presents the results obtained from the design and behavioral simulation of the proposed Traffic Light Controller implemented using Verilog Hardware Description Language (HDL). The controller was developed using a modular architecture based on a Moore Finite State Machine (FSM) and verified through behavioral simulation using EDA Playground with the Icarus Verilog simulator.

The results presented in this chapter include functional verification of the controller, simulation waveform analysis, and validation of different operating modes including normal traffic sequencing, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation. These results demonstrate the correctness and reliability of the proposed traffic control system.

5.2 Simulation Environment

The proposed Traffic Light Controller was verified through behavioral simulation using EDA Playground. The Icarus Verilog simulator was used for compiling and executing the Verilog HDL modules, while EPWave was used to analyze the generated simulation waveforms.

Table 5.1: Simulation Environment

Parameter	Specification
Simulation Platform	EDA Playground
Simulator	Icarus Verilog 12.0
Waveform Viewer	EPWave
Hardware Description Language	Verilog HDL
Controller Architecture	Moore Finite State Machine (FSM)
Design Methodology	Modular RTL Design
Clock Input	50 MHz (System Clock)
Clock Enable	10 Hz Tick Generation
Operating Modes	Normal, Pedestrian, Emergency, Night Mode

5.3 Functional Verification

Functional verification is performed to ensure that the proposed Traffic Light Controller operates correctly under various traffic conditions and operating modes. The design is implemented using Verilog Hardware Description Language (HDL) and verified through behavioral simulation using EDA Playground with the Icarus Verilog simulator. A dedicated testbench applies different input conditions, including normal traffic operation, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode activation. The generated waveforms are analyzed using EPWave to verify correct state transitions, traffic signal sequencing, timer operation, and synchronization of external inputs. The simulation results confirm that the controller responds correctly to all operating conditions while maintaining safe and conflict-free traffic signal operation.

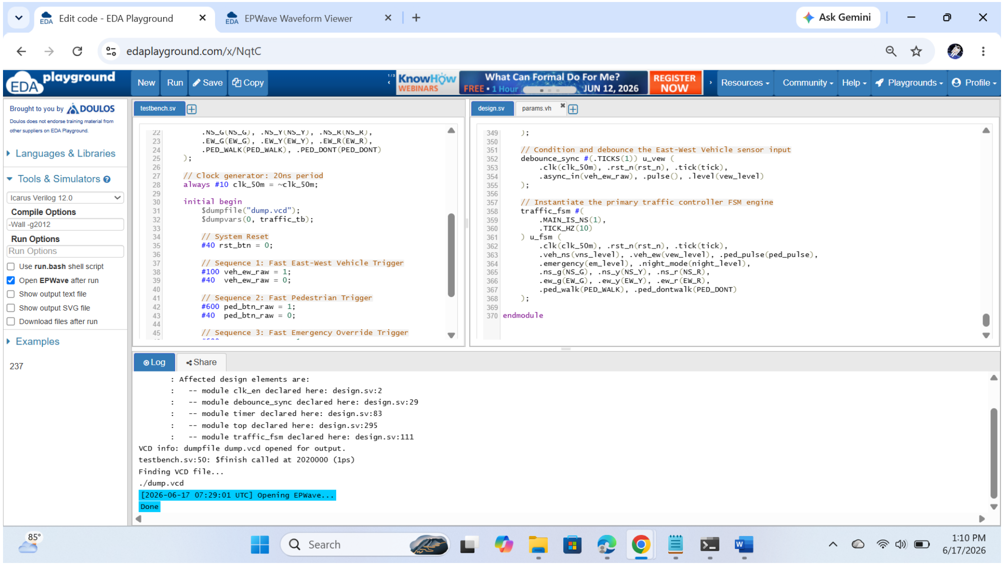

Figure 5.1: Clean Compilation and Runtime Logs

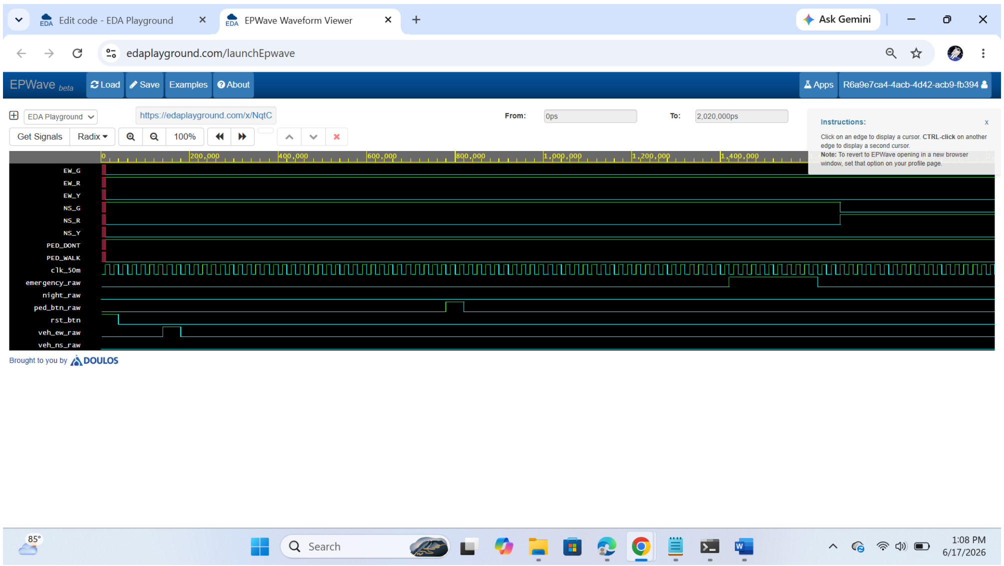

Figure 5.2: Time-Domain Waveform Trace

5.4 Simulation Results

Behavioral simulation was performed using EDA Playground with the Icarus Verilog 12.0 simulator to verify the functionality of the proposed Traffic Light Controller. The generated waveforms were analyzed using EPWave to validate the sequence of traffic signal transitions and the response of the controller under different operating conditions.

The simulation verified the following functionalities:

- Correct sequencing of North–South and East–West traffic signals during normal operation.

- Proper transition through green, yellow, and all-red safety states.
- Successful handling of pedestrian crossing requests through dedicated pedestrian walk and pedestrian flash states.
- Immediate activation of the emergency override mode, forcing all traffic directions into a safe all-red state.
- Correct operation of the night mode, providing blinking traffic signals during low-traffic conditions.
- Reliable processing of asynchronous inputs through synchronization and debounce logic.

The simulation waveforms confirmed that all state transitions occurred according to the designed Moore Finite State Machine without conflicting traffic signals.

5.5 Discussion

The behavioral simulation results demonstrate that the proposed Traffic Light Controller performs correctly under both normal and special operating conditions. The modular architecture simplifies the design by separating the controller into independent functional modules including the clock-enable generator, timer, debounce and synchronization logic, finite state machine, and top-level integration.

The implementation successfully supports pedestrian request handling, emergency vehicle override, vehicle detection, and night mode operation while maintaining deterministic state transitions through the Moore FSM architecture. The use of synchronized and debounced external inputs improves system reliability by preventing false triggering caused by switch bouncing and asynchronous signals.

Overall, the proposed controller provides a reliable and flexible traffic management solution that can be extended for future hardware implementation and intelligent traffic control applications.

5.6 Summary

This chapter presented the behavioral simulation results of the proposed Traffic Light Controller implemented using Verilog HDL. Functional verification confirmed the correct operation of the controller during normal traffic sequencing, pedestrian crossing, emergency vehicle override, vehicle detection, and night mode operation. The generated simulation waveforms validated the correctness of the finite state machine and demonstrated the reliability of the modular controller architecture. The results indicate that the proposed design provides an effective solution for digital traffic signal control and serves as a strong foundation for future hardware implementation and advanced traffic management systems.

CHAPTER 6

Conclusions and Future Scope

6.1 Introduction

This chapter presents the overall conclusions drawn from the work carried out in this project and discusses possible directions for future improvements. The project focused on the design and behavioral simulation of a parameterized Traffic Light Controller using Verilog Hardware Description Language (HDL). The proposed controller was developed using a Moore Finite State Machine (FSM) architecture to provide reliable and efficient traffic signal control for a four-way road intersection.

Throughout this work, the principles of digital traffic control systems, Finite State Machines, synchronization techniques, and modular hardware design were studied in detail. The proposed controller incorporates several practical traffic management features, including pedestrian crossing requests, emergency vehicle override, vehicle detection support, and night mode operation. A modular architecture consisting of clock-enable generation, timer control, debounce and synchronization logic, finite state machine control, and top-level integration was developed to simplify implementation and verification.

The complete design was implemented using Verilog Hardware Description Language and functionally verified through behavioral simulation using EDA Playground with the Icarus Verilog simulator. The simulation results demonstrate that the proposed controller correctly manages traffic signal sequencing under various operating conditions while maintaining safe and conflict-free traffic flow.

6.2 Summary of Work

The primary objective of this project was to design and verify a reliable Traffic Light Controller capable of managing traffic flow at a four-way intersection using a Moore Finite State Machine. The implementation focused on providing a modular, parameterized, and easily maintainable digital controller capable of supporting multiple traffic management features.

The major stages involved in this work include the following:

- **Study of Traffic Light Control Systems**

  A detailed study was carried out on conventional traffic signal systems, digital controller design, and Finite State Machine (FSM) architectures. The operating principles of Moore FSMs and their suitability for sequential control applications were analyzed.

- **Literature Review**

  Existing research related to digital traffic light controllers, FSM-based control systems, and intelligent traffic management techniques was reviewed. The study highlighted the advantages of modular hardware design and the importance of reliable input processing for real-time traffic control applications.

- **Identification of Research Gaps**

  The literature survey identified several limitations in conventional traffic controllers, including fixed timing operation, limited support for pedestrian and emergency conditions, and inadequate handling of asynchronous external inputs. These observations motivated the development of a modular traffic controller supporting multiple operating modes.

- **System Design**

  A Moore Finite State Machine architecture was developed to control the sequence of traffic signals. The controller was designed to support normal traffic operation, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode functionality.

- **Hardware Architecture Design**

  A modular hardware architecture was developed consisting of a clock-enable generator, debounce and synchronization module, timer module, finite state machine controller, and top-level integration module. This modular approach simplifies system verification and future enhancements.

- **Verilog HDL Implementation**

  The proposed controller was implemented using Verilog Hardware Description Language. Individual RTL modules were developed independently and integrated to form the complete Traffic Light Controller.

- **Simulation and Functional Verification**

  The functionality of the proposed controller was verified through behavioral simulation using EDA Playground with the Icarus Verilog simulator. Simulation waveforms were analyzed using EPWave to validate correct traffic signal sequencing, pedestrian request handling, emergency override, vehicle detection, and night mode operation.

6.3 Key Observations

Several important observations were made during the design and behavioral simulation of the proposed Traffic Light Controller.

First, the Moore Finite State Machine (FSM) provides a reliable and systematic approach for implementing sequential traffic control logic. Since the output signals depend only on the current state, the controller produces stable and deterministic traffic signal transitions without unwanted glitches.

Second, the modular architecture simplifies the design by separating the controller into independent functional modules, including the clock-enable generator, debounce and synchronization logic, timer module, finite state machine controller, and top-level integration. This modular organization improves code readability, simplifies debugging, and allows future system enhancements.

Third, reliable handling of asynchronous external inputs is essential for real-world traffic control applications. The synchronization and debounce circuits successfully eliminate switch bouncing and reduce the possibility of metastability, ensuring that only valid input events influence the controller operation.

Finally, the implementation using Verilog Hardware Description Language together with behavioral simulation in EDA Playground demonstrates that complex traffic management logic can be efficiently modeled, verified, and validated before hardware implementation.

6.4 Advantages of the Proposed System

The proposed Traffic Light Controller offers several advantages compared to conventional fixed-time traffic signal systems.

- **Modular Design**

  The controller is divided into independent functional modules, making the design easier to understand, verify, maintain, and extend.

- **Reliable Traffic Signal Control**

  The Moore Finite State Machine ensures deterministic state transitions and prevents conflicting traffic signal combinations, thereby improving traffic safety.

- **Support for Multiple Operating Modes**

  The controller supports normal traffic sequencing, pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation within a single architecture.

- **Reliable Input Processing**

  Synchronization and debounce circuits ensure accurate processing of asynchronous external inputs, reducing false triggering caused by switch bouncing and metastability.

- **Configurable Timing**

  The parameterized design allows traffic signal timing to be modified according to different intersection requirements without changing the overall controller architecture.

- **Easy Verification**

  The modular Verilog implementation enables straightforward behavioral simulation and functional verification using EDA Playground, Icarus Verilog, and EPWave.

6.5 Limitations of the Current Work

Although the proposed Traffic Light Controller performs correctly during behavioral simulation, several limitations remain in the current implementation.

First, the controller has been verified only through behavioral simulation. Hardware implementation on an FPGA development board would provide additional validation under real operating conditions.

Second, the controller currently follows predefined timing parameters and does not dynamically adjust signal durations according to real-time traffic density.

Third, the present implementation does not include communication interfaces for integration with centralized traffic monitoring or intelligent transportation systems.

Finally, advanced optimization techniques such as adaptive traffic control and machine learning based traffic prediction are beyond the scope of the current work.

6.6 Future Scope

Several improvements can be explored in future work to further enhance the proposed Traffic Light Controller.

6.6.1 FPGA Hardware Implementation

The proposed controller can be implemented on an FPGA development board such as the Digilent Nexys A7 to evaluate real-time operation, hardware resource utilization, and system performance.

6.6.2 Adaptive Traffic Control

Future versions of the controller can dynamically adjust traffic signal timing based on vehicle density measured using multiple traffic sensors.

6.6.3 IoT-Based Traffic Monitoring

The controller can be integrated with Internet of Things (IoT) technologies to enable remote monitoring, centralized traffic management, and real-time data collection.

6.6.4 Artificial Intelligence Integration

Machine learning algorithms may be incorporated to analyze traffic patterns and automatically optimize signal timing for improved traffic flow.

6.6.5 Smart City Applications

The proposed controller can be integrated with smart city infrastructure to coordinate multiple traffic intersections and improve overall transportation efficiency.

6.6.6 Enhanced Emergency Vehicle Management

Future work may include wireless communication between emergency vehicles and traffic controllers to automatically provide priority traffic clearance.

6.7 Conclusion

This project presented the design and behavioral simulation of a parameterized Traffic Light Controller using Verilog Hardware Description Language based on a Moore Finite State Machine architecture. The controller successfully manages traffic signal sequencing while supporting pedestrian crossing requests, emergency vehicle override, vehicle detection, and night mode operation through a modular hardware design.

Behavioral simulation using EDA Playground with the Icarus Verilog simulator verified the correct operation of the controller under various traffic conditions. The modular implementation, synchronization logic, timer module, and finite state machine architecture provide a reliable and scalable foundation for digital traffic control systems.

The proposed controller demonstrates that Verilog HDL can be effectively used to design and verify reliable traffic management systems. With future FPGA implementation and the integration of adaptive traffic management, IoT connectivity, and intelligent decision-making techniques, the proposed design can be further extended into an advanced smart traffic control system suitable for modern transportation applications.

Mapping of Program Outcomes (POs) and Sustainable Development Goals (SDGs)

The proposed project titled *"Design and Simulation of a Parameterized Traffic Light Controller Using Verilog HDL"* contributes to the attainment of several Program Outcomes (POs) defined for the Electronics and Communication Engineering program. The project involves the application of digital system design, finite state machine implementation, hardware description language programming, behavioral simulation, and modular system development for intelligent traffic management applications.

PO1: Engineering Knowledge:
Applied the principles of digital electronics, sequential logic design, finite state machines, and Verilog HDL to develop a reliable Traffic Light Controller.

PO2: Problem Analysis:
Analyzed the limitations of conventional fixed-time traffic signal systems and identified the need for a modular controller supporting pedestrian requests, emergency vehicle priority, vehicle detection, and night mode operation.

PO3: Design/Development of Solutions:
Designed and developed a parameterized Traffic Light Controller using a Moore Finite State Machine with modular architecture for reliable and efficient traffic signal management.

PO4: Conduct Investigations of Complex Problems:
Verified the functionality of the controller through behavioral simulation by testing different traffic scenarios including normal operation, pedestrian crossing, emergency override, vehicle detection, and night mode.

PO5: Engineering Tool Usage:
Utilized Verilog Hardware Description Language for RTL design and EDA Playground with Icarus Verilog and EPWave for behavioral simulation and waveform analysis.

PO6: The Engineer and the World:

Developed a traffic management solution that contributes toward safer transportation systems by improving traffic control and reducing the possibility of conflicting traffic movements.

PO7: Ethics:

Maintained accuracy and integrity throughout the project by validating the design through simulation and presenting only verified results.

PO8: Individual and Collaborative Team Work:

Participated in the design, coding, testing, debugging, and integration of individual modules while effectively collaborating with team members throughout the project.

PO9: Communication:

Prepared comprehensive documentation, simulation reports, block diagrams, state diagrams, and technical presentations to clearly communicate the design methodology and project outcomes.

PO10: Project Management and Finance:

Planned and executed the project systematically by dividing it into design, implementation, verification, and documentation phases while utilizing available software resources efficiently.

PO11: Life-Long Learning:

Enhanced knowledge of digital system design, finite state machine implementation, Verilog HDL programming, and traffic management systems through continuous learning and practical implementation.

2. Program Specific Outcomes (PSOs)

PSO1: Ability to design and implement digital systems using modern hardware design tools

The project demonstrates the application of Verilog HDL and finite state machine design principles to implement a modular Traffic Light Controller. The design illustrates the ability to develop reliable digital hardware systems using industry-relevant design methodologies.

PSO2: Application of Electronics and Embedded System Concepts in Practical Applications

The project applies digital design concepts to solve a real-world traffic management problem. The modular architecture provides a foundation for future FPGA implementation and integration with intelligent transportation systems.

3. Alignment with Sustainable Development Goals (SDGs)

The proposed Traffic Light Controller indirectly contributes to several Sustainable Development Goals by promoting safer transportation, efficient infrastructure, and sustainable urban development.

SDG 11: Sustainable Cities and Communities

Efficient traffic signal management improves urban mobility, reduces traffic congestion, enhances road safety, and contributes toward sustainable transportation systems.

SDG 3: Good Health and Well-Being

Improved traffic regulation helps reduce road accidents and promotes safer travel for both motorists and pedestrians.

Overall Impact

The proposed Traffic Light Controller demonstrates the practical application of digital system design principles for solving real-world traffic management problems. The modular Moore Finite State Machine architecture enables reliable traffic signal sequencing while supporting pedestrian requests, emergency vehicle priority, vehicle detection, and night mode operation.

The implementation using Verilog HDL and behavioral simulation demonstrates how digital hardware can be utilized to develop dependable control systems for intelligent transportation applications. The project also provides a scalable platform that can be extended with FPGA implementation, adaptive traffic control, IoT connectivity, and artificial intelligence for future smart city applications.

Overall, this work highlights the importance of digital hardware design, modular architecture, and finite state machine based control systems in developing reliable and efficient traffic management solutions.

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